



Sep 3 2026

Sales Data Analysis & Visualization Using Python

Presented by

Amogh.R.Hanmattekar

Introduction

Sales data analysis is the process of examining sales information to understand business performance, customer demand, and profitability. In this project, Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn were used to clean, analyze, and visualize the sales dataset. The analysis helps identify important sales patterns and supports better business decision-making.

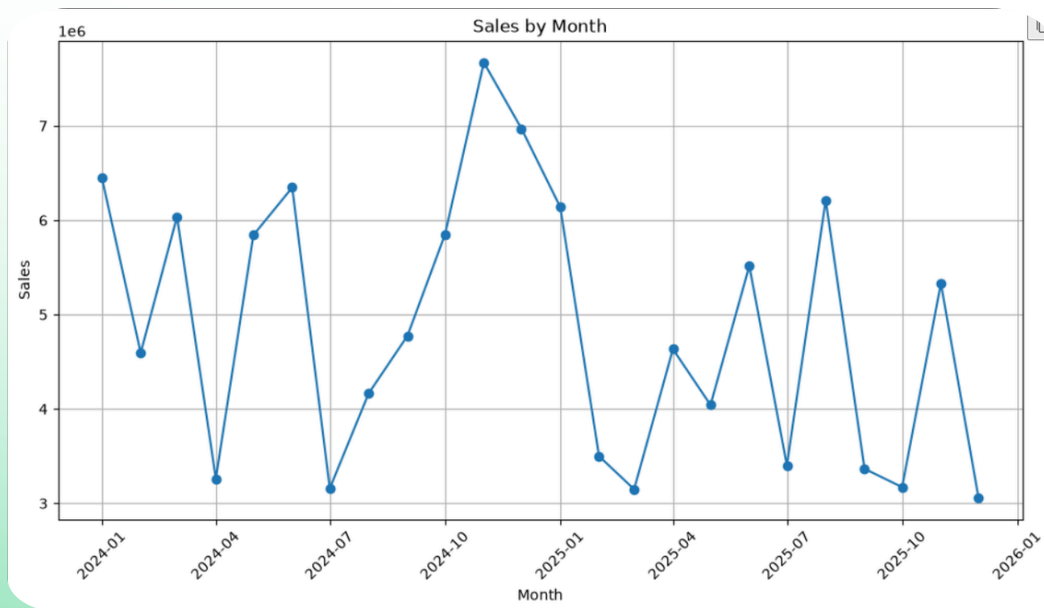


Table of Contents

MAJOR FINDINGS	1
BEST PERFORMING CATEGORY	2
BEST PERFORMING REGION	3
MOST PROFITABLE PRODUCTS	4
SALES TRENDS	5
DICSOUNT/PROFIT OBSERVATION	6
BUSSINESS RECOMENDATIONS	7

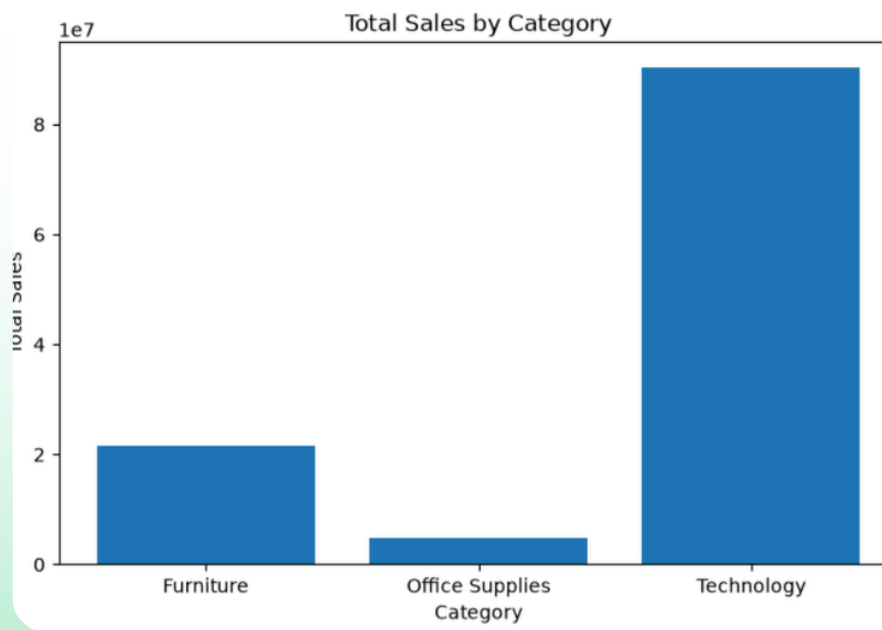
MAJOR FINDINGS

The analysis provided insights into sales, profit, categories, regions, products, discounts, and sales trends. It helped identify high-performing areas as well as products and factors affecting profitability.



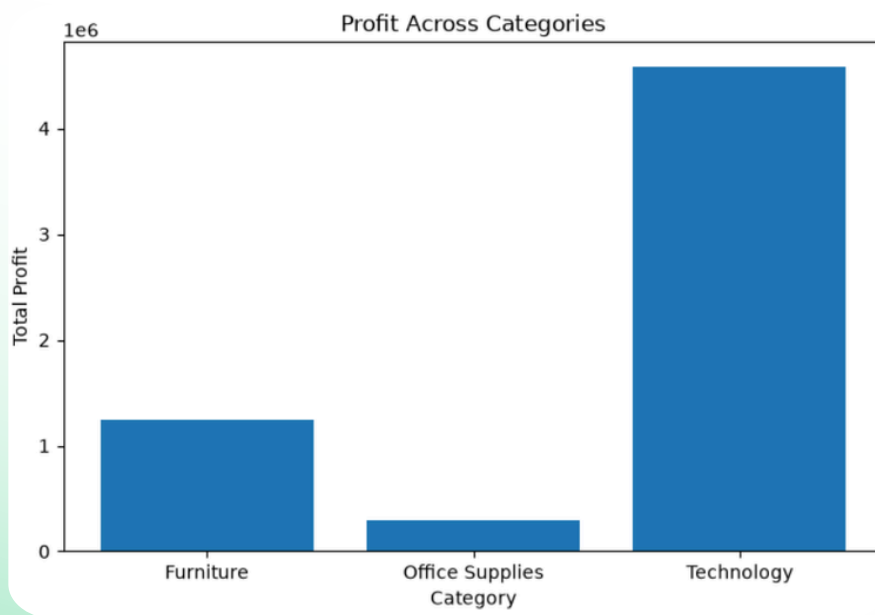
Best Performing Category

Technology was identified as the best-performing category based on total sales. It contributed significantly to overall business revenue.



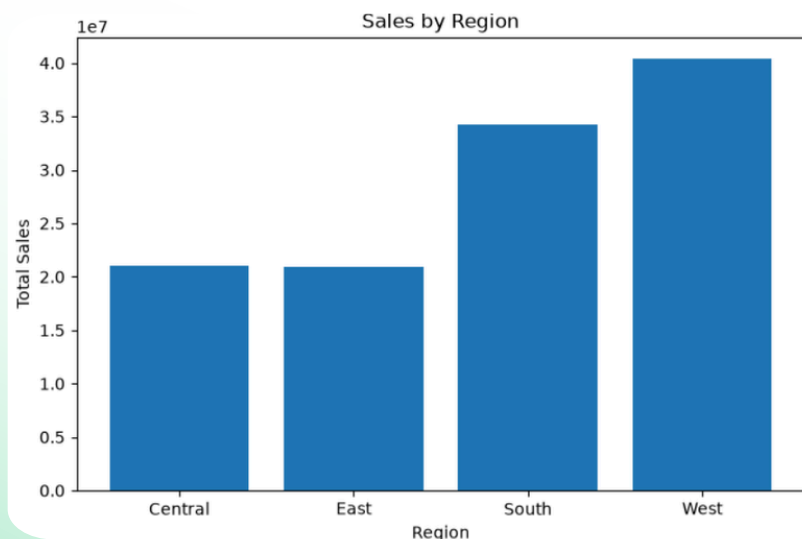
Best Performing Region

The West region was identified as the best-performing region based on total sales, indicating strong customer demand and business performance in this region.



Most Profitable Products

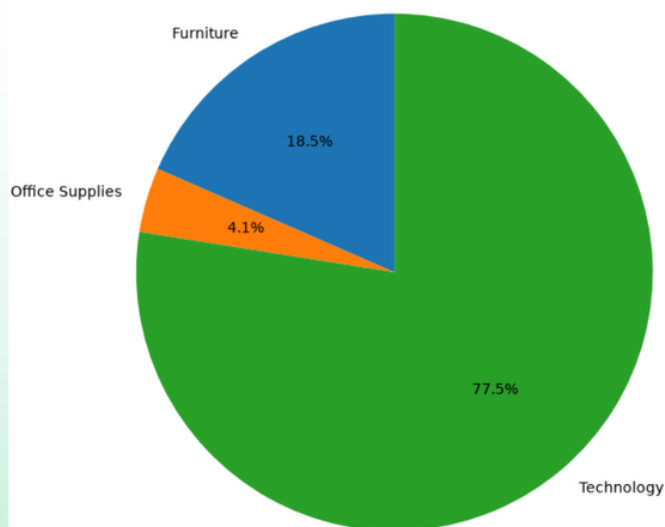
The analysis identified the products generating the highest profit. These products are important contributors to overall profitability and should be given greater attention in future sales strategies.



Sales Trends

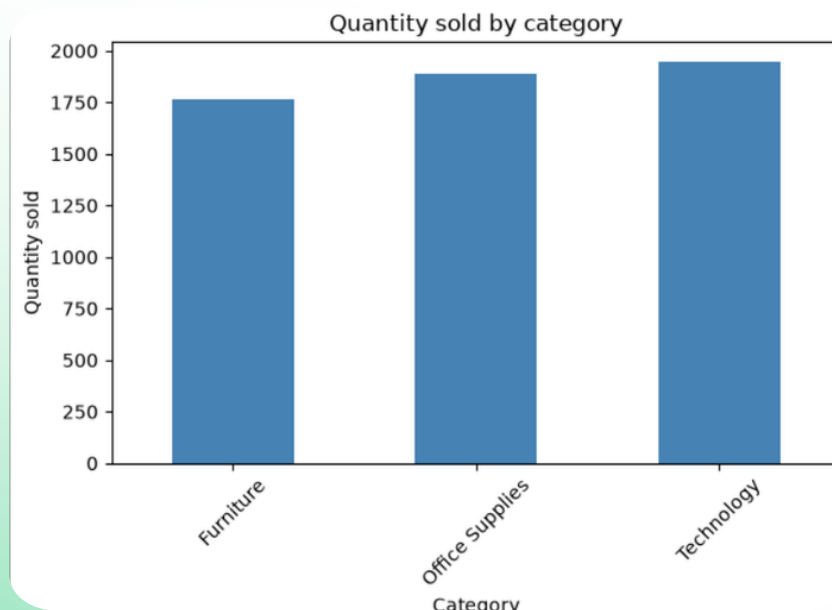
The analysis of sales over time showed changes and patterns in sales performance. Visualizations helped identify periods of higher and lower sales, which can be useful for planning inventory and marketing activities.

percentage contribution of each category to total sales



Discount/Profit Observation

Higher discounts do not always result in higher profits. Excessive discounts can reduce profit margins, especially for products that already have low profitability. Therefore, discounts should be carefully planned.



Business Recommendations

- Focus on high-performing categories and products.
- Improve sales strategies in weaker regions.
- Control excessive discounts to protect profit margins.
- Promote highly profitable products.
- Use sales trends for better inventory and marketing planning.
- Continue using data analysis to support business decisions.

Conclusion

The Sales Data Analysis and Visualization project successfully analyzed sales data using Python. The analysis provided useful insights into category performance, regional sales, product profitability, sales trends, and the impact of discounts on profit. Data visualization made the results easier to understand and compare. Overall, the findings can help businesses improve sales strategies, control costs, focus on profitable products, and make better data-driven decisions.



Thank you any...?

Q...

Presented by

Amogh.R.Hanmattekar